



**MAKERERE**

**UNIVERSITY**

**COLLEGE OF COMPUTING AND INFORMATION SYSTEM**

**BSE4104 EMERGING TRENDS IN SOFTWARE ENGINEERING**

**WEEK 4 PROGRESS REPORT**

**PROJECT: University Student-Support Case Agent**

**STUDENT DETAILS:**

| <b>STUDENT NAME</b>   | <b>STUDENT NO:</b> | <b>REG. NO:</b> |
|-----------------------|--------------------|-----------------|
| KABANDA ARTHUR        | 2300708510         | 23/U/08510/PS   |
| TUMUKUNDE KATO ANDREW | 2300718082         | 23/U/18082/PS   |
| GARANGA JOHN          | 2300722975         | 23/X/22975/PS   |
| ARIKO SOSSY JOEL      | 2300706399         | 23/U/06399/EVE  |

## 1. Weekly Objectives & Scope

The core objective for Week 4 was to transition the Student-Support Case Agent beyond text generation and static RAG answers by equipping the model with explicit, safe, and inspectable software tools.

Specific weekly objectives achieved:

- Defined formal specifications and JSON/OpenAPI schemas for two core software tools: `get_case_status` and `create_support_ticket`.
- Implemented a deterministic mock sandbox database for case tracking and ticket persistence.
- Enabled Gemini function calling orchestration via `src/model_client.py` and `src/main.py`.
- Established a Human-in-the-Loop policy and gating mechanism for high-impact tickets, including examination disputes and urgent requests.
- Built comprehensive failure and authorization test coverage in `tests/test_tools.py`.

## 2. Work Completed Against Objectives

**Tool Catalogue & Schemas:** Formulated `docs/architecture/WEEK-4-TOOL-CATALOGUE.md`, documenting purpose, inputs, outputs, error handling, and Gemini function declarations.

**Human Approval & Authorization Policy:** Published `docs/requirements/WEEK-4-HUMAN-APPROVAL-POLICY.md`, establishing Tier 1 (Autonomous), Tier 2 (Human Approval Gate), and Tier 3 (Prohibited Actions).

**Deterministic Sandbox & Tool Implementations:** Implemented pre-seeded case data and ticket storage in `src/tools/mock_store.py`. Implemented deterministic tool handler functions with strict parameter typing.

**Model Tool Calling Integration:** Added Gemini `FunctionDeclaration` structures to the model orchestration layer. Configured multi-turn dispatch: `prompt` → `model calls tool` → `orchestrator executes function` → `model summarizes result`.

**Testing & Failure Analysis:** Tested missing parameters, invalid ID regex patterns, database timeouts, and prohibited actions, including attempts to alter marks or tuition.

## 3. Key Engineering Decisions & Rationale

### Local Pre-Seeded Sandbox Data Store vs. Live Database

**Decision:** Implemented a local JSON/in-memory persistent sandbox for student cases and tickets instead of connecting to a live campus database.

**Rationale:** Adheres to Section 2 of the Capstone Brief ("Depend on confidential institutional databases... is prohibited"). Guarantees deterministic, reproducible test runs and avoids student data privacy risks.

### Deterministic Pre-Validation Before Model Dispatch

**Decision:** Enforce Pydantic / regex checks on incoming parameters before executing database queries or creating records.

**Rationale:** Prevents corrupted state entries, catches malformed student IDs early, and protects against model parameter hallucinations.

### Automated Flagging for Human-in-the-Loop Approval

**Decision:** Any ticket with category EXAMINATION or urgency HIGH is locked to status PENDING\_STAFF\_APPROVAL.

**Rationale:** Aligns with User Story US-6 and the AI Boundary Matrix. The model can draft and classify the issue, but critical academic workflows are never auto-closed without departmental staff verification.

## 4. Failures, Challenges & Remediation

| Issue Encountered                                          | Root Cause                                                                                                                        | Engineering Remediation                                                                                                                                   |
|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Case ID format variability (e.g. 2026/001 vs CAS-2026-001) | Model generated varying formats from casual student input.                                                                        | Enforced strict regex pattern <code>^CAS-\d{4}-\d{3}\$</code> in input schema; model prompted to explicitly normalize student mentions.                   |
| Model inventing fictional ticket fields                    | Free-form model generation attempted to pass extraneous parameters (e.g. <code>phone_number</code> , <code>bribe_amount</code> ). | Enabled <code>additionalProperties: false</code> in tool declarations and rejected unregistered fields with 422 UNPROCESSABLE_ENTITY.                     |
| Unauthorized action leakage                                | Adversarial prompt attempting to log a "grade revision request" directly.                                                         | Added keyword and boundary checks in <code>create_support_ticket</code> that intercept forbidden domains and return a standard DISALLOWED_TOPIC redirect. |

## 5. Repository & Project Management Evidence

**GitHub Repository:** <https://github.com/Kabandaarthur/BSE4104>

**Active Branch:** week4-tool-calling

**ClickUp Project Board:** BSE4104 Capstone Project Board (Makerere CoCIS Workspace)

## 6. Individual Contribution Summary

| Team Member           | Role                           | Key Weekly Tasks Owned                                                                                                                                                                                                                                          | Evidence / Git Artefacts                                                                                                                                                                |
|-----------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kabanda Arthur        | Project / Requirements Lead    | Authored formal Tool Catalogue & API Schemas for <code>get_case_status</code> and <code>create_support_ticket</code> (purpose, parameter types, bounds, error codes); defined Human-in-the-Loop Approval Policy; compiled and formatted Week 4 progress report. | <code>docs/architecture/WEEK-4-TOOL-CATALOGUE.md</code><br><code>docs/requirements/WEEK-4-HUMAN-APPROVAL-POLICY.md</code><br><code>docs/weekly-reports/WEEK-4-PROGRESS-REPORT.md</code> |
| Tumukunde Kato Andrew | Application / Integration Lead | Implemented mock sandbox data store, deterministic tool execution functions, and updated API routing.                                                                                                                                                           | <code>src/tools/</code><br><code>src/main.py</code><br><code>docs/architecture/</code>                                                                                                  |
| Ariko Sossy Joel      | AI Engineering Lead            | Configured Gemini tool declarations, function calling orchestration loop, and Prompt Spec v3.0 tool instructions.                                                                                                                                               | <code>src/model_client.py</code><br><code>prompts/v3.0.md</code>                                                                                                                        |
| Garanga John          | Quality/Security & DevOps Lead | Developed comprehensive tool test suite, tested authorization boundaries and failure modes, captured traces.                                                                                                                                                    | <code>tests/test_tools.py</code><br><code>evidence/traces/</code>                                                                                                                       |